Two-mode Network Analysis of Culture Concepts

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Abstract

Kroeber and Kluckhohn's classic 1952 monograph *Culture: A Critical Review of Concepts and Definitions* [Kroeber and Kluckhohn, 1952] brought together over a hundred scholarly definitions of culture and indexed the key concepts they discuss. We build one network from that index to explore how concepts and definitions are connected. In the *definitions network*, concepts like "society" and "beliefs" are linked to 171 definition codes, grouped into seven categories and credited to the author or authors that K&K index under each. Each code is one quotation, identified by its author, year, page numbers and opening words. We present basic statistics about the network, visualize the data using correspondence analysis and spanning trees, and highlight the key concepts and definitions that anchor it. Data-driven types found in the wording of the definitions do not recover K&K's categories at better than chance.

1 Data and networks

We build one network from the Index B of Kroeber and Kluckhohn [1952], hereafter K&K. The *definitions network* links 470 concepts—concrete ideas like "society," "customs," and "beliefs"—to 171 definitions of culture, which K&K number as codes. Each code is credited to the author or authors K&K index under it, and falls into one of seven categories, which K&K label with the capital letters A through G: A Descriptive, B Historical, C Normative, D Psychological, E Structural, F Genetic, and G Incomplete.[1] Of the 171 codes, 169 appear in K&K's own author index. The other two, C-I-18a and D-II-16, are attributions cited in the main text of the book that the index does not list. Boroch [2016] quotes two parts of four K&K codes separately, lettered (a) and (b): A-3, E-5, F-II-4 and F-II-7. We keep each of these as two codes, A-3(a) and A-3(b) and so on, which is why the eight variant codes replace four. K&K print one index entry per code, so each variant starts with its parent's concept links. For 29 of the 41 concept links copied in this way, we kept the link only on the variant whose quotation contains the concept's name; the other 12 remain on both variants. That assignment is our reading of the quotation text, not taken from anything directly printed in the K&K index. Concepts have consistent names; where a concept appears under different headings in K&K's source text, we treat it as a single concept. Each code is one quotation, identified by its author, year, page numbers and opening words. Section A of the appendix describes how each code was matched to its quotation and how the a and b variants were built.

The 171 × 470 definitions network naturally breaks into 10 separate components, with the largest containing 626 nodes. Our analysis focuses on this largest component, which contains virtually all the meaningful connections. We use maximum spanning trees across multiple projections to identify the strongest connections in different views of the network.

[1] K&K further divide three of the seven categories into numbered subcategories: Normative (C) into C-I and C-II, Psychological (D) into D-I through D-IV, and Genetic (F) into F-I through F-IV. The other four categories have no subcategories. Codes that sit in a subcategory carry its numeral, as in C-I-18a or F-II-4 below.

2 Overview of the network

Table 1 gives summary statistics for the definitions network. The first three rows show node and edge counts. The "degree" rows count connections of each node type; a concept's degree is how many definitions mention it, while a definition's degree is how many concepts it contains. Concepts and definitions differ substantially in their average degree.

Table 1: Summary statistics for the definitions network. Statistics describe the full network, before the restriction to the largest component's 2-core used in the analyses. Average nearest-neighbor degree (k_{nn}) is the average degree of a node's neighbors; degree correlation shows how a node's degree relates to the degree of its neighbors.

	Definitions network
Concept nodes	470
Definitions	171 codes (5 isolated)
Edges	1201
Density	0.015
Mean degree, concepts	2.56
Minimum degree, concepts	1
Maximum degree, concepts	36
Mean degree, definitions	7.02
Minimum degree, definitions	0
Maximum degree, definitions	25
Mean k_{nn}, concepts	10.18
Mean k_{nn}, definitions	9.22
Degree correlation r	-0.076

The degree correlation is -0.076: concepts with many connections tend to link to definitions with few, and vice versa, though the association is weak. The nearest-neighbor rows show the other side of this pattern. A concept's neighboring definitions contain 10.2 concepts on average, and a definition's neighboring concepts appear in 9.2 definitions on average, so concepts attach preferentially to heavily used definitions and definitions attach to widely shared concepts.

Figure 1 shows the degree distributions for the concept nodes (left) and the definition nodes (right). Most concepts are lightly connected: 79% appear in two or fewer definitions. Definitions are more heavily connected, with the average definition covering 7 concepts (median 6) and a few covering many more, up to 25. This asymmetry means that a few concepts are discussed widely, while most concepts appear in just one or two definitions.

3 Maximum spanning trees

By construction, the two-mode definitions network has no ties between two concepts or between two definitions, so we analyze the implied concept-to-concept ties and the defintions-to-definitions connections via multiple one-mode projections [Everett and Borgatti, 2013]. Following Breiger [1974], two concepts are tied if they share at least one definition, with weight equal to the number they share. Likewise, two definitions are tied through the concepts they share. In matrix form, the projections are:

$$W_{\text{concept}} = AA^{\top}, \qquad W_{\text{def}} = A^{\top}A,$$

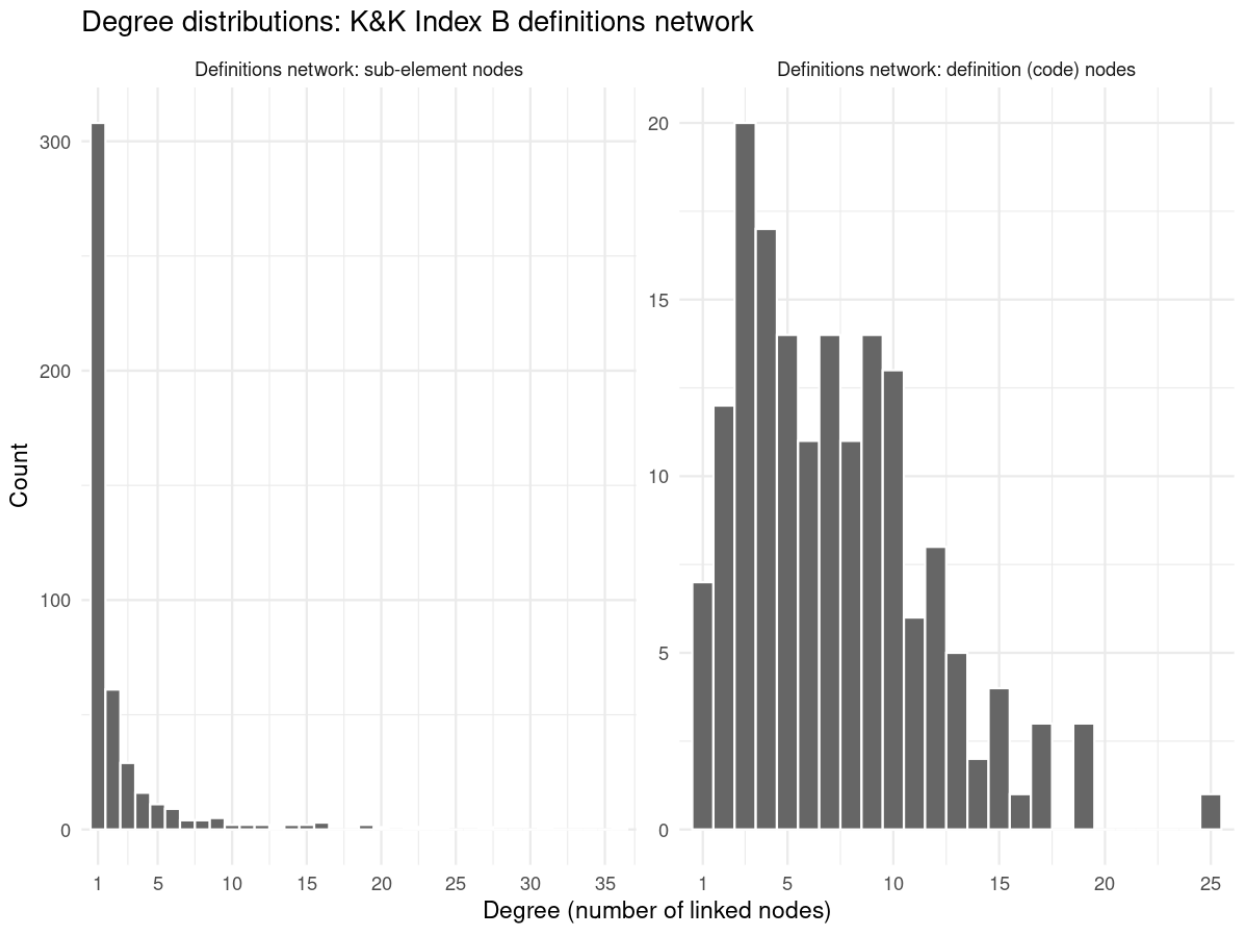


Figure 1: Degree distributions of the definitions network. Left: sub-element (concept) nodes. Right: definition (code) nodes.

where $(W_{\text{concept}})_{ij} = \sum_k a_{ik}a_{jk}$ counts the definitions that include both concepts i and j. Each projection is a weighted graph whose edges are the off-diagonal pairs with positive weight. We build projections from the 2-core of the network's largest component.

Projected graphs are typically dense and fully connected making it hard to detect their underlyinng structure. To do so, we extract the "backbone" [Neal, 2014] of each projection by extracting the maximum spanning tree of each projection. The maximum spanning tree of each projection is found by computing the minimum spanning tree on negated weights, $-w$. Kruskal [1956] sorts edges from largest weight to smallest and adds each unless it would close a cycle, producing a tree that maximizes $\sum_{e \in T} w_e$. Because the 2-core is connected, each tree has exactly $|V| - 1$ edges. In the visualizations, node size shows betweenness within the tree and node color shows the community structure detected using Newman's leading-eigenvector method [Newman, 2006] applied to the tree.

The definitions network has two main projections. The concept-level tree (Figure 2) shows how 160 concepts cluster. The definition-level tree (Figure 3) shows how 149 definitions cluster. Table 2 lists the top 10 concepts and definitions by betweenness in their maximum spanning trees.

In the concept tree (Figure 2), the six largest nodes—group, customs, social, behavior, society, and beliefs—are the top six of Table 2. Group sits at the center of a pink region of communication

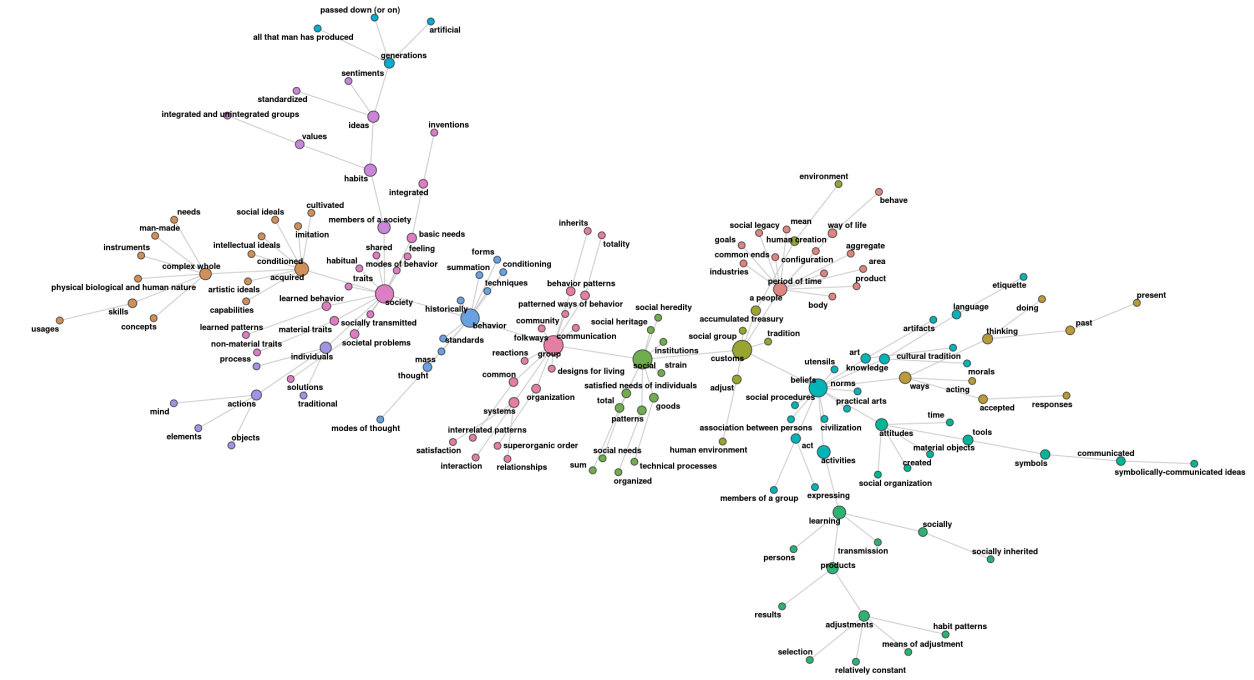


Figure 2: Maximum spanning tree of the definitions network, sub-element (concept) nodes. Color is the leading-eigenvector community on the tree; size is betweenness in the tree.

and organization vocabulary (communication, organization, folkways, systems, designs for living). Social heads a green region of social-organizational vocabulary (social heritage, institutions, social needs, patterns, goods), and customs heads an olive region of tradition and adjustment (social group, accumulated treasury, adjust, association between persons). Behavior and society head the blue and magenta regions on the left, which hold the historical and learned-behavior vocabulary (historically, standards, learned behavior, socially transmitted). Beliefs heads the teal region of norms, knowledge and artifacts on the right, with norms, practical arts, act and civilization. A peripheral branch below beliefs runs through activities and learning to products and adjustments. This chain collects the acquisition and transmission vocabulary of the Genetic definitions and hangs off the core through a single path, so these ideas connect to the rest of the network only transitively. At the top left, ideas, habits, values and generations form a second branch that joins society through habits.

The definition tree (Figure 3) puts its three largest nodes, C-I-6, F-I-19, and A-16, at the center of the tree, with C-I-6 the largest of all. C-I-6 heads the largest community (24 definitions, drawn from all seven groups), and A-16 sits in the same community. F-I-19 and A-17 share a second community at the left of the center. This is consistent with the table, where five of the top ten definitions carry a C or F code and the others are Descriptive, Historical or Psychological. Degree and betweenness dissociate in this ranking: C-I-6 has both the highest betweenness and the highest tree degree (17), acting as the main hub of its community, while F-I-19 has the second-highest betweenness with only eight ties, sitting on the paths that join its cluster to the rest of the tree. One author supplies a striking share of the bridges, as Bidney holds three of the ten most central definitions (A-16, C-II-4, and C-II-3). Away from the center, the Descriptive definitions A-3(a), A-3(b), A-5, A-10a, and A-12 form a separate olive branch in the upper right, and the Descriptive

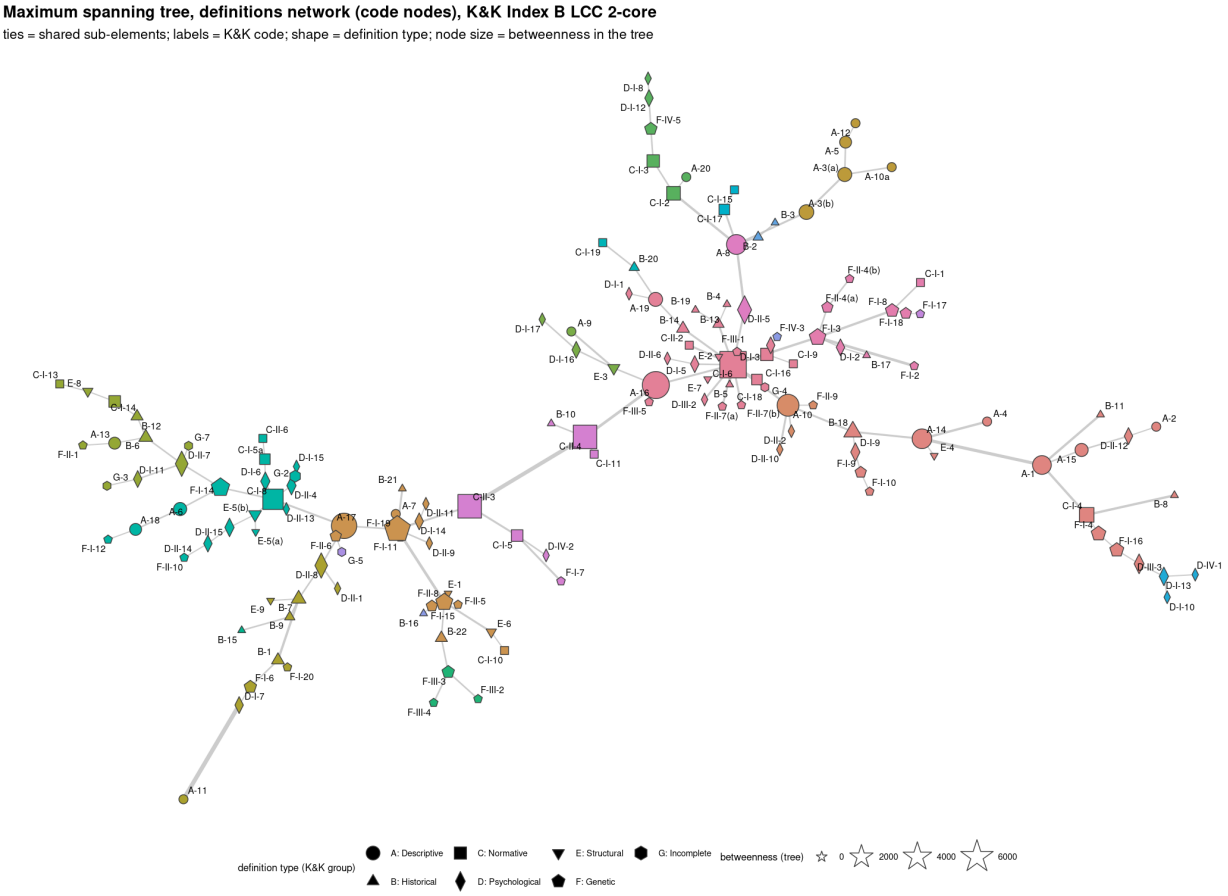


Figure 3: Maximum spanning tree of the definitions network, code (definition) nodes. Color is the leading-eigenvector community on the tree; size is betweenness in the tree.

and Historical definitions A-1 and A-14 run along the right side, showing that the earliest and simplest definitions connect to the core through fewer, longer paths. Figure 4 repeats the definition tree with each code labeled by the author and year of its quotation and shaped by the publication type of the work K&K index under it; repeated labels are different quotations by the same author in one year.

The author–year view puts names to the central codes. The hub C-I-6 is the quotation from Gillin and Gillin's 1942 book, the second-most central node is Murdock's 1949a article (F-I-19), and Bidney holds the next three—the articles of 1947, 1946 and 1942 behind A-16, C-II-4 and C-II-3. Kroeber 1948a is the only other work with two codes in the top ten (A-17 and B-18), and Linton holds two (the 1945b chapter C-I-8 and the 1936 book A-10). The works themselves barely cluster. The 149 codes of the tree carry 120 distinct author–year labels, and the 19 labels that appear two or more times almost never stay together: 15 of them spread over two or more communities. Kluckhohn and Kelly's 1945a chapter, the most quoted work in the corpus with seven of its eight codes in the tree, reaches four communities scattered across the tree, and Herskovits 1948 likewise spans four. The three a/b quotation pairs do sit together, one pair per community (Dixon 1928, Blumenthal 1937, Feibleman 1946), as those variants share most of their concept links; Linton 1945a is the partial exception, with its two E-5 variants together and its third code (C-I-9) in another community. Publication type cuts across this structure. Books dominate the tree (97 of the 149 nodes), but no community consists of a single publication type, so the regions of the tree organize

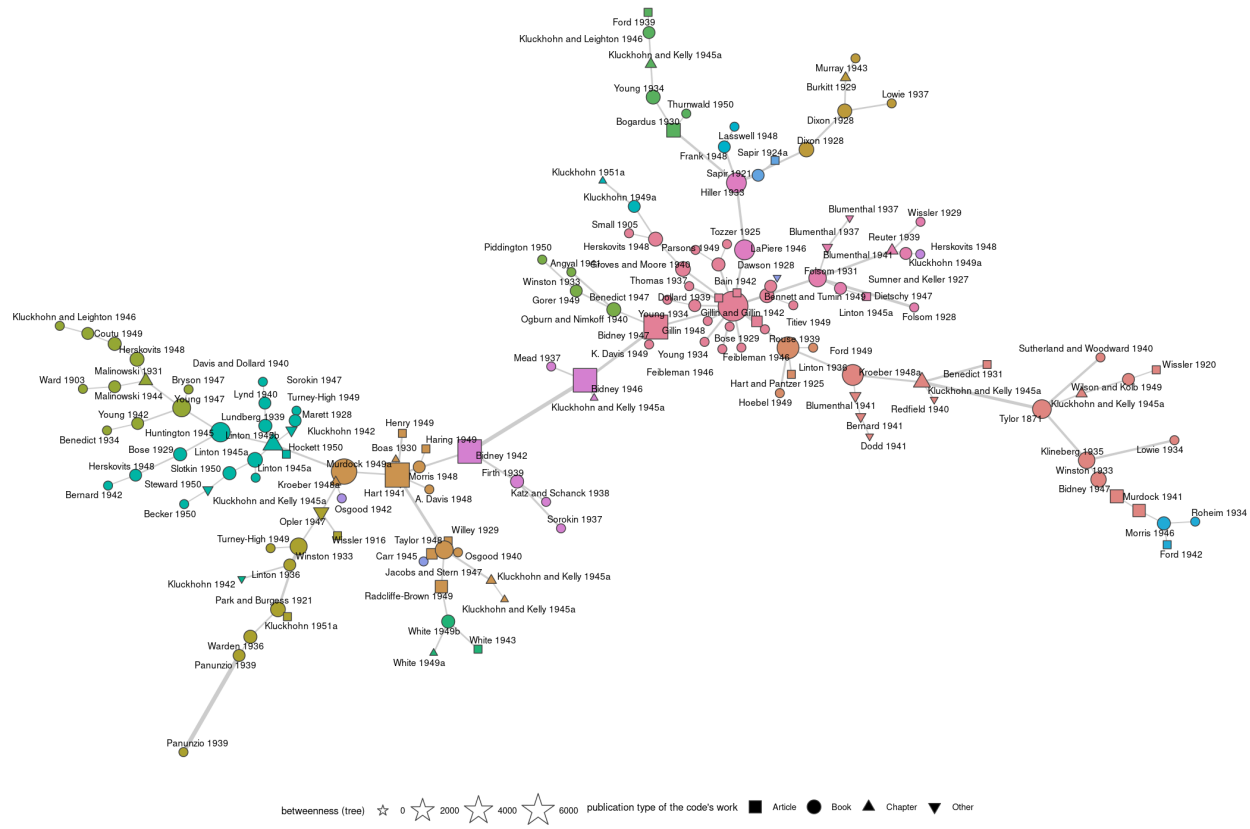


Figure 4: Maximum spanning tree of the definitions network, code (definition) nodes labeled by the author and year of their quotation. Shape is the publication type of the work K&K index under the code; color is the leading-eigenvector community on the tree; size is betweenness in the tree.

concept vocabulary, not the kind of work the definitions were drawn from.

The six most central concepts, group, customs, social, behavior, society and beliefs, are the top of Table 2, and each anchors one of the color regions of the concept tree. The most central definition is C-I-6, at 71% of the maximum possible, a single Normative code. In the concept tree the acquisition and transmission vocabulary sits on peripheral branches rather than at the core, suggesting that Genetic concerns, however common in the definitions themselves, connect to the rest of the conceptual structure only through a narrow path.

4 Data-driven types: non-negative matrix factorization

The spanning trees and the community colors above classify nodes hard: each code belongs to exactly one community of the tree. Non-negative matrix factorization (NMF) [Lee and Seung, 2001] offers a soft, parts-based alternative. It approximates a non-negative matrix V by a product WH with all entries non-negative, so every code carries a loading on each factor and every factor is an additive mixture of the original features, with no subtractive terms. We ask how far data-driven types of the definition wording agree with K&K's own seven categories A–G. The factorization uses the same base-R implementation throughout (Euclidean multiplicative updates, deterministic

Table 2: Top 10 nodes by Freeman-normalized betweenness in the definitions network MST.

Node type	Node	Norm. Betw. (%)	Degree
Concepts	group	59.39	12
	customs	58.04	7
	social	57.38	10
	behavior	50.83	10
	society	50.66	14
	beliefs	48.40	13
	acquired	17.81	9
	a people	15.72	13
	activities	14.04	2
	learning	13.22	5
Definitions	C-I-6	71.45	17
	F-I-19	55.57	8
	A-16	53.42	4
	C-II-4	50.94	4
	C-II-3	50.88	3
	A-17	43.88	4
	C-I-8	32.15	6
	A-10	27.01	5
	B-18	22.97	3
	D-II-5	19.42	2

Note: Definition codes use the format X-Y-Z, where X is a letter (A–G) for K&K's seven definition categories (Descriptive, Historical, Normative, Psychological, Structural, Genetic, Incomplete), Y is a roman numeral for the subcategory, and Z is the number of the definition within it. Variants such as A-3(a) and A-3(b) are the two quotations of one K&K code, and they are counted as separate codes (see the data section).

given a seed, with ten random restarts per value of k), and it fixes k at seven, one factor per K&K group. Fixing k this way is a choice, not a result: the reconstruction error falls monotonically in k without an elbow, and restart agreement is irregular (mean adjusted Rand index 0.48 to 0.70 over $k = 3$ to 8), so neither criterion selects a number of factors on its own.

4.1 Types from the wording of the definitions

The analysis asks the question of the definition texts themselves, independently of K&K's coding. The corpus holds 171 definition codes carrying a quotation: the Table 2 texts mapped to K&K codes through the crosswalk, the appendix-only texts, and one statement K&K prints itself (a Part II page footnote). Three Table 2 columns that repeat another column's quotation word for word are not counted again. Documents average about thirty-four words (median 28), so the term–document matrix is built restrictively: tokens of three letters or more, a stopword list that removes the words nearly every definition contains (culture, society, social, man, and the general definitional vocabulary total, whole, learned, acquired), and terms kept to a document frequency between two documents and half the corpus, leaving 403 terms weighted by TF–IDF. The four quotations K&K and Boroch print in French or German enter as English translations (K&K's own rendering for Maquet, ours for the other three, not yet checked against the printed pages). The factorization is run at $k = 7$, one factor per K&K group.

The factors again find recognizable themes rather than K&K's categories (Figure 5), and we

label them by their leading terms: *species-specific ability* (events, species, use, dependent, ability), *order and operation* (order, operation, self, superorganic, complex), *mental and holistic* (concept, mean, holistic, traits, mental), *goals and persons* (certain, rather, others, persons, goals), *symbolic process and adjustment* (concrete, process, symbolic, experience, adjust), *material and transmitted behavior* (material, non, transmitted, socially, institutions), and *passed down generations* (generation, passed, down, everything, common). On a corpus this small a factor can be driven by one author's phrasing, so the labels should be read as descriptive of the top terms, not as clean concepts. The labels for mental and holistic and goals and persons are the least clear, since their leading terms include generic words (rather, certain, some, mean). Against the A–G groups the agreement is close to chance (adjusted Rand index 0.012, against a permutation 95th percentile of 0.016), and no factor has more than 54% of its codes in one group.

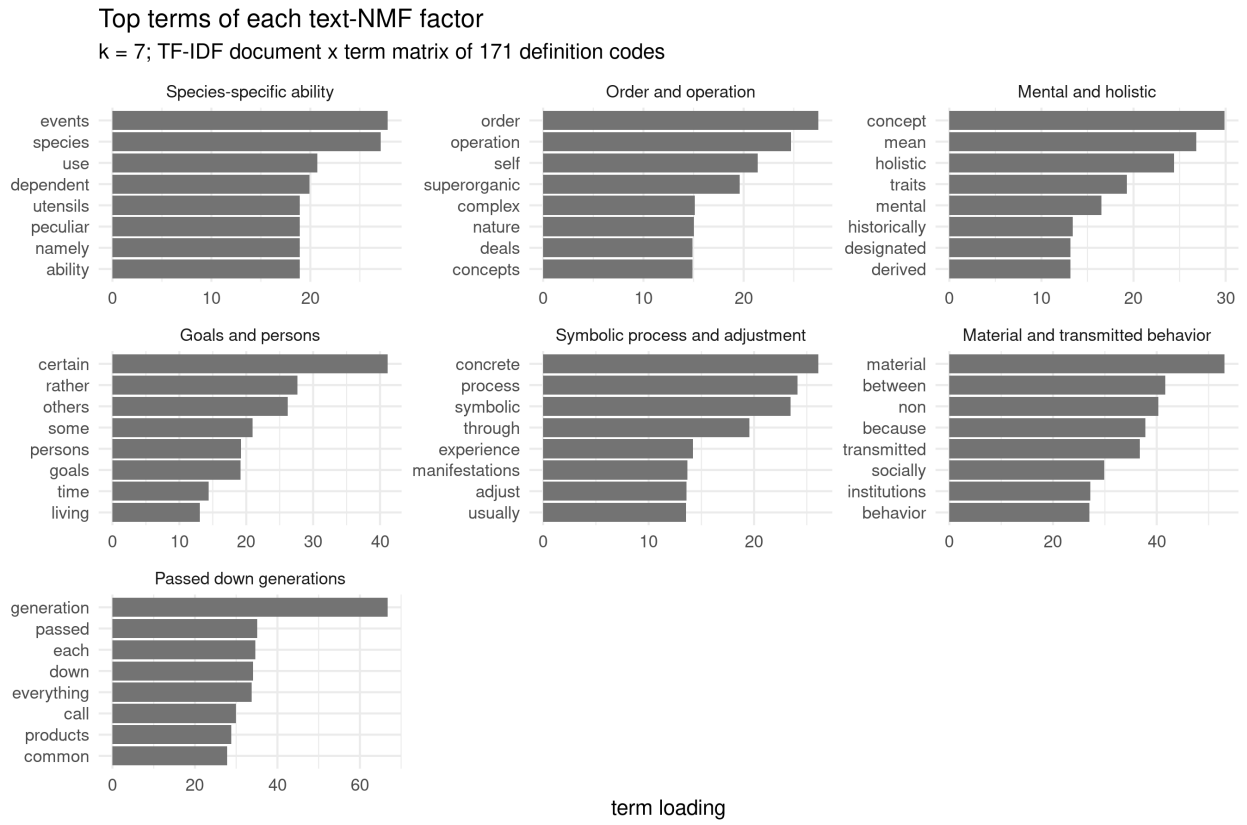


Figure 5: Top terms of each factor of the text NMF at $k = 7$, from the TF–IDF matrix of the 171 definition codes.

4.2 Interpretation

The wording of the definitions does not recover K&K's taxonomy. Two readings are possible. One is that K&K's groups classify the emphases of individual *statements*, and that these emphases cannot be recovered from quotations of about thirty words each. The other, which we find more plausible, is that the seven groups are editorial categories—useful for organizing a bibliography—rather than dimensions along which the corpus varies. This is consistent with the spanning-tree picture above, where the vocabulary the Genetic group concerns itself with sits on a peripheral chain of the concept tree rather than forming a region of its own, and with the community structure of the trees, which

follows concept vocabulary, not authorial schools. The null result should be read against the method's limits as well: the number of factors was fixed to match the taxonomy under test rather than selected by the data, the short corpus makes TF–IDF factors fragile (restart agreement ranges from 0.48 to 0.70 for k from 3 to 8), and the quotations are the passages Boroch selected, not the authors' full works.

5 Limitations

The networks depend on the K&K Index B transcription and on the verified-code filter. Element names are not unique across headings, so same-named elements are merged, which can join unrelated entries. Several readings depend on OCR corrections recorded in the repository's manual-fix files; the validators check structure and round-trips, not the content of each edge. Two codes that are not in Index A are included: C-I-18a and D-II-16, from the main text. K&K's Part III statements (C-5, D-5, D-7, E-17) are excluded. The basis for each is documented in the repository. The four split codes and the 29 link decisions that separate their variants (see the data section) are judgments: the variants share one K&K index entry, and the text-based assignment of concept links is a reading that a reader can check against the quotations, not a printed fact. The definition-level trees and the text factors depend on these decisions.

A Constructing the code-level data

The analyses use one object: a table with a row for each of the 171 K&K definition codes. Each row gives the code's group and subgroup, its sub-elements, and the citation that identifies its quotation. This section records how that table was built, the choices made along the way, and the basis for each. Two sources are joined. K&K's Index A and Index B, read from a scan of the 1952 monograph, give each code its author, year and sub-elements. Boroch's article Boroch [2016] gives each code a quotation with page numbers. Every join follows a stated rule, and each judgment is recorded with its basis in the project's provenance log.

A.1 Sources and the unit

The unit is the code: one printed index entry in K&K, and one quotation in Boroch. Boroch's appendix has 171 codes. His Table 2 has 135 columns, 132 of which match an appendix code. The two sources number codes differently. Boroch renumbers within each group, so his A10 (Winston 1933) is K&K's A-9. Codes are therefore matched by author, year, group and subgroup. Positional agreement is kept only as a check: of 100 matches made by name and year, 48 agree with the position.

A.2 Index A and Index B

Index A (authors) and Index B (sub-elements) were read from the OCR text layer and corrected by explicit rules. The corrections are recorded in a manual-fix table of 142 rows, each with its printed-page evidence. A further 10 rows recover codes that the OCR glued into element names, and 14 rows record name decisions. The project avoids general character rules, because a rule such as reading a y as a 5 silently produced codes from unrelated tokens. Each manual fix is keyed to a parsed token, and the checks confirm that every key matches one.

The universe is 171 codes. Of these, 169 appear in K&K's author index. The other two, C-I-18a (Maquet 1949) and D-II-16 (Aberle et al. 1950), are cited in main-text footnotes but not listed

in the index; their Index B entries are read from the sub-element entries that carry them. Four Part III statement codes (C-5, D-5, D-7 and E-17) are excluded, since they are K&K's statements rather than Part II definitions.

A.3 Matching Boroch's codes to K&K's

The crosswalk matches each Table 2 column to a K&K code. Of the 135 columns, 119 match automatically when exactly one code fits. The remaining 16 are settled by 15 hand decisions, each with a basis, and one recorded slip correction. Three Table 2 columns have no appendix entry of their own: A10a, C1-5a and F4-6. Each is a verbatim copy of another quotation. A10a repeats A10 (Winston 1933), C1-5a repeats C1-5 (Firth 1939), and F4-6 repeats F4-5, which is the entry Boroch's numbering misprints as F4-6. None of the three adds a quotation, so the object does not use them. The F4-5 quotation is Kluckhohn and Kelly 1945a (printed p. 87), and it gives K&K's F-IV-5. K&K's printed codes A-10a (Lowie 1937) and C-I-5a (Lynd 1940) take their quotations from the appendix entries A12 and C1-6.

A.4 Quotations and citation keys

Each code has one quotation. Boroch supplies 170 of them: the 132 Table 2 codes and the 39 appendix-only codes, less the Sears addendum (C1-22), which is linked to a specialty code rather than to a network code. D-III-4 has no Boroch entry, and its quotation is K&K's printed footnote 14 (Roberts 1951, pp. 3 and 6, printed p. 60). Four quotations are not in English and are rendered from a translation table: B-17 and C-I-18a from French, and F-I-5 and F-II-3 from German. C-I-18a uses K&K's own printed English. The other three are working translations that have not been checked on the page images.

Each code is identified by a citation key: Boroch's author, year and page numbers, together with the first six words of the quotation. The key is unique for all 171 codes, with no missing values. The page numbers are Boroch's as printed and have not been checked against the source works. The key therefore identifies a quotation as Boroch reports it, and the page numbers are the first thing to check against a source.

A.5 Splitting the a and b quotations

Boroch quotes parts of four K&K codes separately, labeled (a) and (b). K&K prints one index entry for each of these codes, so the object had one code where Boroch has two quotations. On 2026-10-30 the project split each of the four codes into two variants (Table 3). The split follows Boroch's quotations and is not a distinction K&K prints. Each variant takes its own quotation and its own citation key.

Because K&K gives one index entry per code, each variant starts with a copy of its parent's sub-element links, 41 in all. A lexical screen then assigns 29 of the 41 to one variant. A link stays on the variant whose quotation contains the sub-element's name and not on the other. The 12 links where both or neither quotation contains the name remain on both variants. The screen is a word match, so a name used in a generic sense, such as "total" or "past," can be assigned wrongly. Each assignment is a reading of the quotation text and should be treated as a judgment. Removing a link also removes the heading edge that carries it, which accounts for 34 edge rows. Together the split takes the verified network from 167 to 171 codes and from 1189 to 1201 edges.

Table 3: The four split codes and their variants.

K&K code	Variant	Boroch code and source	Page numbers
A-3	A-3(a)	A3, Dixon 1928	3
A-3	A-3(b)	A4, Dixon 1928	3
E-5	E-5(a)	E5, Linton 1945a (footnote 5)	5
E-5	E-5(b)	E6, Linton 1945a	32
F-II-4	F-II-4(a)	F2-4, Blumenthal 1937 (footnote 4)	3
F-II-4	F-II-4(b)	F2-5, Blumenthal 1937	12
F-II-7	F-II-7(a)	F2-8, Feibleman 1946 (tentative; footnote 7)	75
F-II-7	F-II-7(b)	F2-9, Feibleman 1946 (final)	76

A.6 Checks and open items

The build's validators check the network's structure, its round trips and its attributes, but not the content of each edge. The block check places 1301 of the 1326 Index B edge rows (before the 34-row removal) in their printed block, and 25 in another block. The page-image review covers 23 of those 25. Several items remain open. The year of K&K's footnote 5, which Boroch gives as Linton 1945a, was read from the page image at 150 dpi, and a higher-resolution check is still needed. The three duplicate columns need recorded decisions in the crosswalk, and the text analysis must be brought into line with them. The three non-English translations need checking against the page images. The Boroch page numbers need checking against the source works. The 29 link assignments are judgments on the text, as described above.

B Supplementary sankey diagrams at the subgroup level

The NMF comparison in the body classifies the definition wording against K&K's seven group letters. The figure here repeats it at the finer grain of K&K's own typology, the roman-numeral subgroups the classification prints within groups C, D, and F; groups A, B, E, and G number their codes plainly and have no subgroup level. Each ribbon takes the color of its origin node's group, and node size is the number of codes.

On the left, A, C-I and D-II reach all seven factors, and B, D-I, F-I and F-II each reach six. On the right, *material and transmitted behavior* is the largest factor at 43 codes and draws on D (10), B (7) and F (7). *Passed down generations* (34 codes) draws most on D (11) and A (8), and *symbolic process and adjustment* (26 codes) on F (12 codes, 46% of the factor), B (5) and D (4). *Order and operation* (21 codes) draws on F (7), D (4), and A, B and C (3 each). *Species-specific ability* and *mental and holistic* (17 codes each) both draw on F (6) and C (5). *Goals and persons* (13 codes) is mostly D (7 codes, 54% of the factor).

For the definition codes no aggregation is needed, since each code carries its own subgroup. The largest cell holds eight codes (A on *passed down generations*, 8 of 22 codes), and two cells hold seven (F-I on *symbolic process and adjustment*, 7 of 21; B on *material and transmitted behavior*, 7 of 23). No subgroup's codes sit on a single factor. The smallest subgroups, C-II (6 codes), D-III (4) and D-IV (2), reach between two and four factors.

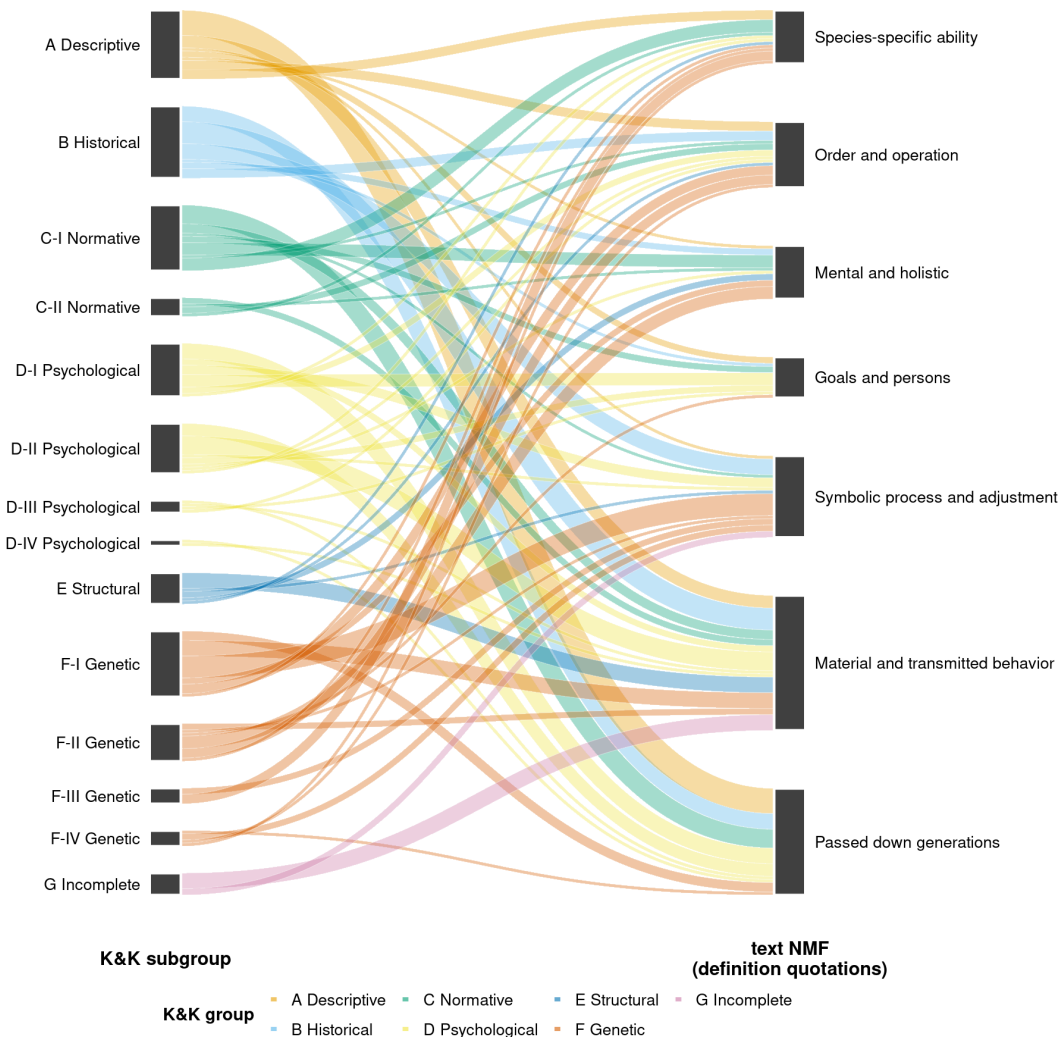


Figure 6: The 171 definition codes carrying a text, by their own K&K subgroup (left) and dominant text NMF factor (right). Ribbons take the color of their origin node's group.

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